

Project Report On,

Statistical Analysis of Automobile Companies

Trading in National Stock Exchange

Subject: Financial Mathematics

Subject Code: QF2025203

SUBMITTED BY:

Arsh Nayak	PGDMQF-004
Komal Sankhla	PGDMQF-012
Lehar Parikh	PGDMQF-013
Mahima Jain	PGDMQF-014
Pratibha Jain	PGDMQF-017

**Under the Guidance of,
Prof. Anju Pandey**

January- 2026

Table of Contents

Introduction of TATA Motors	4
Introduction of Maruti Suzuki	5
Introduction of Mahindra & Mahindra	6
Introduction of Nifty 50.....	7
Methodology	8
1. Data Collection:	8
2. Tool Used:.....	8
3. Formulae:	8
Daily Stock Price of TATA Motors	10
Monthly Stock Price of TATA Motors	12
Yearly Stock Price of TATA Motors	14
Daily Stock Price of Maruti Suzuki	15
Monthly Stock Price of Maruti Suzuki	17
Yearly Stock Price of Maruti Suzuki.....	19
Daily Stock Price of Mahindra & Mahindra Ltd	20
Monthly Stock Price of Mahindra & Mahindra Ltd	22
Yearly Stock Price of Mahindra & Mahindra Ltd	24
Daily Stock Price of NSE Index	25
Monthly Stock Price of NSE Index	27
Yearly Stock Price of NSE Index	29
Daily return comparison of the companies.....	30
Monthly return comparison of the companies	32
Annually return comparison of the companies.....	34
Daily descriptive statistics of the company:.....	35
Monthly descriptive statistics of the company	36
Yearly descriptive statistics of the company:.....	37
Correlation.....	38
Daily Correlation Analysis: Automobile Companies vs Nifty 50	38
Monthly Correlation Analysis: Automobile Companies vs Nifty 50	39
Yearly Correlation Analysis: Automobile Companies vs Nifty 50	40
Regression	41
Daily price	41
TATA Motors.....	41
Maruti Suzuki.....	42
Mahindra & Mahindra.....	43
Monthly price	45
TATA Motors.....	45
Maruti Suzuki.....	46

Mahindra & Mahindra.....	48
Yearly price.....	50
TATA Motors.....	50
Maruti Suzuki.....	51
Mahindra & Mahindra.....	52
Interpretation.....	53
Overall Interpretation	54
Conclusion	55
References.....	56



Introduction of TATA Motors

Industry/Sector & Headquarters:

- Tata Motors operates in the **automobile sector**, manufacturing commercial vehicles (CV), passenger vehicles (PV), and electric vehicles.
- Headquartered in Mumbai, India.

Key Operations:

- The group includes Tata Motors, Tata Passenger Vehicles (TMPV), Tata Commercial Vehicles, and luxury subsidiary Jaguar Land Rover (JLR).
- Offers a wide range of vehicles: cars, trucks, buses, electric vehicles, and premium segment vehicles.

FY 2024-25 Financial Performance:

- **Revenue (Consolidated):** ₹4,39,695 crore (up 1.3% YoY).
- **EBITDA:** ₹57,600 crore.
- **Profit Before Tax (PBT, bei):** ₹34,330 crore.
- **Net Profit:** ₹28,100 crore.

Key Milestones & Stock-impacting Events:

- Ongoing demerger expected by mid-FY26, creating separate listed entities for CV and PV segments, including merger of TMPV and Jaguar Land Rover into the main entity.
- Commercial Vehicle (CV) division unveiled 14 new products at Bharat Mobility Global Expo 2025, across powertrains from diesel to hydrogen combustion engines.
- Continued leadership in SUV and EV market share, crossing 200,000 cumulative EV sales.
- Approval and execution of merger between Tata Motors Finance and Tata Capital to focus on core emerging tech and vehicle segments.

Introduction of Maruti Suzuki

Industry/Sector & Headquarters:

- Maruti Suzuki operates in the **automobile sector** (passenger vehicles) and is headquartered in New Delhi, India.

Key Operations:

- The company leads India's passenger vehicle segment, manufacturing hatchbacks, sedans, SUVs, and exporting vehicles to global markets.
- Maruti Suzuki is the top exporter among Indian automakers, shipping nearly 43% of India's total passenger vehicle exports in FY25.

FY 2024-25 Financial Performance:

- **Net Sales:** ₹1,45,115 crore (up 7.5% YoY).
- **Net Profit:** ₹13,955 crore (up 5.6% YoY).
- **Annual Sales:** 2,234,266 units (domestic: 1,901,681; exports: 332,585).
- **Market Capitalization:** Approximately ₹4,80,000 crore.

Key Milestones & Stock-impacting Events:

- Record sales and exports for second consecutive year, crossing 2 million sales milestone.
- Strong growth in exports, particularly with launches of the Fronx and Jimny 5-door models exported to Japan.
- Announced commencement of exports for Suzuki's first global strategic BEV (e VITARA) at the Bharat Mobility Global Auto Expo 2025.
- Launch of the all-new Victoris SUV, strengthening its SUV portfolio.
- Strategic sales to Toyota Kirloskar Motor doubled YoY, aiding financial growth.

Introduction of Mahindra & Mahindra

Industry/Sector & Headquarters:

- Mahindra & Mahindra operates in the **automobile and farm equipment sector**. Headquartered in Mumbai, India.

Key Operations:

- The group's core businesses include automotive (SUVs, commercial vehicles, electric vehicles) and tractors/farm machinery via Mahindra Tractors.
- Also prominent in financial services, IT (Tech Mahindra), and real estate.

FY 2024-25 Financial Performance:

- **Revenue (Automotive and Farm, Consolidated):** Mahindra Finance net interest income: ₹99,404 million, profit after tax: ₹22,619 million (up 17% YoY).
- **Automotive Milestones:** Notable traction in EV deliveries (over 6,300 e-SUVs delivered).
- **Gross Loan Book (Finance):** Closed at ₹1,06,339 crore.

Key Milestones & Stock-impacting Events:

- Launch of several new models (Thar Roxx, XUV 3XO) and expanded BEV offering.
- Large investment allocation of ₹37,000 crores across Auto, Farm, and Services for FY25-27.
- Strong export growth in the automotive segment, with focus on South Africa and Australia.
- Continued capacity expansion in SUVs, and unveiling of a new platform expected from FY27.

Introduction of Nifty 50

The Nifty 50 is a benchmark stock market index representing the weighted average of 50 of the largest and most liquid Indian companies listed on the National Stock Exchange (NSE). Launched in 1996, it serves as a barometer of the Indian equity market and reflects the overall market sentiment and economic health of the country.

The index covers 13 sectors of the Indian economy, including financial services, information technology, energy, consumer goods, and pharmaceuticals, making it a well-diversified representation of the market. Companies included in the Nifty 50 are selected based on market capitalization, liquidity, and trading frequency, and the index is reviewed semi-annually to ensure it remains relevant and representative.

From a quantitative perspective, the Nifty 50 is widely used for statistical analysis, portfolio optimization, risk assessment, and performance benchmarking. Analysts and researchers apply various quantitative methods such as time series analysis, regression models, and volatility forecasting to study its behavior and derive insights for investment strategies.

In summary, the Nifty 50 is not just a market index but a vital tool for quantitative research, offering a robust dataset for analyzing trends, correlations, and financial dynamics within the Indian stock market.

Methodology

For this analysis project, we studied the stock price performance of Automobile Company like,

1. TATA Motors
2. Maruti Suzuki
3. Mahindra & Mahindra

and compared them with the NSE Sensex (**Nifty Pharma**). The Research Methodology followed into three main steps:

1. **Data Collection,**
2. **Tools Used,** and
3. **Formulas Applied**

1. Data Collection:

- Historical stock price data for were taken from, NSE website: <https://www.nseindia.com/>
- NSE Sensex index data (**Nifty 50**) was also collected from the same source for ensuring consistency.
- Data was collected for three time periods:
 - **Daily Data** – last 30 trading days. (From 28th February, 2025 to 28th March, 2025)
 - **Monthly Data** – last 24 months. (From 03rd April, 2023 to 03rd March, 2025)
 - **Yearly Data** – Last 10 years. (From 01st April, 2016 to 01st April, 2025)

2. Tool Used:

- **Microsoft Excel:** For data storage, calculations (returns, descriptive statistics) and data visualizations (line plots, boxplots).
- **Statistical Functions:** Such as mean, median, standard deviation, skewness and kurtosis were calculated using built-in formulas or statistical libraries.
- **Charts and Graphs:** were prepared using Excel to show trends, returns, histograms, and correlations.
- **Microsoft Word:** For report compilation

3. Formulae:

➤ **Returns:** Percentage change calculated in Excel as:
$$((\text{Current Price} - \text{Previous Price}) / \text{Previous Price}) * 10$$

➤ **Descriptive Statistics:**

- a. Mean: AVERAGE (range)

- b. Median: MEDIAN (range)
- c. Mode: MODE.SNGL(range)
- d. Standard Deviation: STDEV.S(range)
- e. Quartiles: QUARTILE.EXC (range, 1/2/3)
- f. Percentiles: PERCENTILE.EXC (range, k)
- g. Skewness: SKEW (range)
- h. Kurtosis: KURT (range)
- i. Correlation Analysis

Software Used: Microsoft Excel

Function Used:

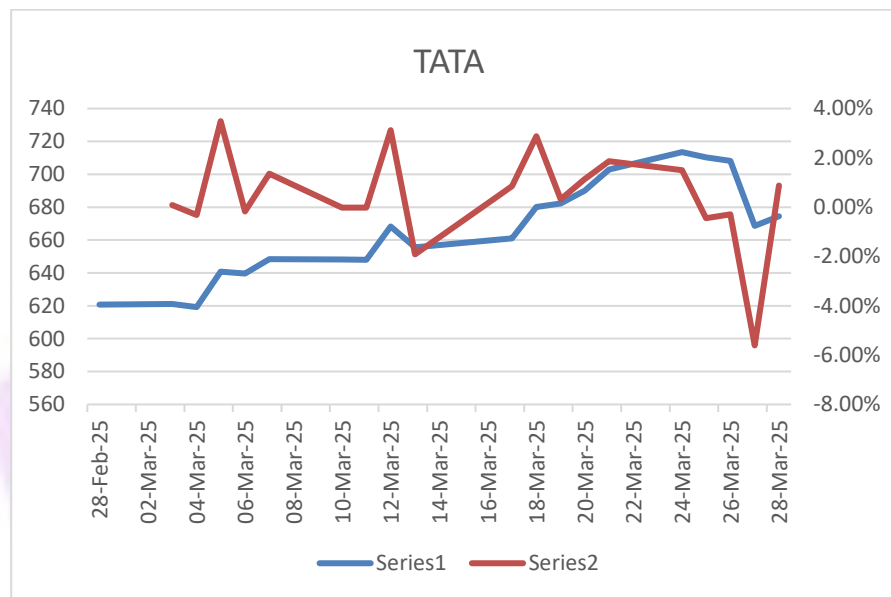
=CORREL(range1, range2)

- j. Regression Analysis

Software Used: Microsoft Excel (Data Analysis ToolPak → Regression)

Daily Stock Price of TATA Motors

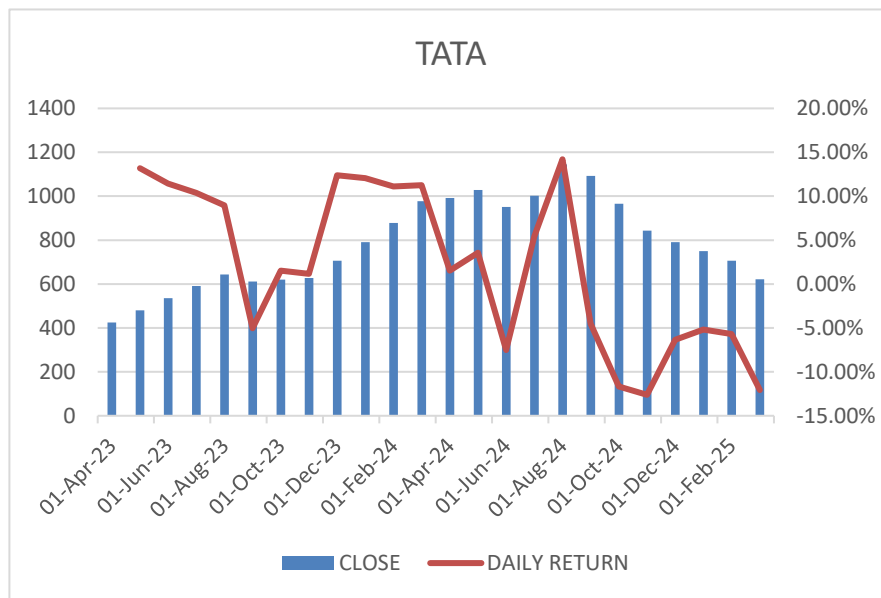
Date	TATA				
	OPEN	HIGH	LOW	CLOSE	DAILY RETURN
28-Feb-25	645	645	618.45	620.65	
03-Mar-25	620.65	626.85	606.3	621.15	0.08%
04-Mar-25	613.45	629.15	613	619.2	-0.31%
05-Mar-25	619	641.6	618.6	640.8	3.49%
06-Mar-25	653.25	655.95	635.6	639.65	-0.18%
07-Mar-25	640	651.5	635	648.3	1.35%
10-Mar-25	648.3	654.25	643.35	648.15	-0.02%
11-Mar-25	640	651.15	638.45	648.05	-0.02%
12-Mar-25	654.5	671.9	652.25	668.3	3.12%
13-Mar-25	670.65	671.85	649.6	655.5	-1.92%
17-Mar-25	665.05	666.45	657.45	661.05	0.85%
18-Mar-25	665	681.7	663.9	680.05	2.87%
19-Mar-25	685.4	685.75	678.55	682.25	0.32%
20-Mar-25	685	692.7	679	690.05	1.14%
21-Mar-25	693	705	688.55	702.95	1.87%
24-Mar-25	710	717.5	702.05	713.5	1.50%
25-Mar-25	719	723.05	706.35	710.3	-0.45%
26-Mar-25	710.3	717.85	707	708.25	-0.29%
27-Mar-25	668	676.6	661.1	668.55	-5.61%
28-Mar-25	671.85	684.95	669.3	674.45	0.88%



The daily return chart for TATA Motors shows a fluctuating trend with mostly positive returns. The variation in returns highlights how the stock performs on a day-to-day basis. Peaks and troughs in the chart indicate periods of gain and loss. Overall, the chart reflects short-term volatility in TATA Motors' stock.

Monthly Stock Price of TATA Motors

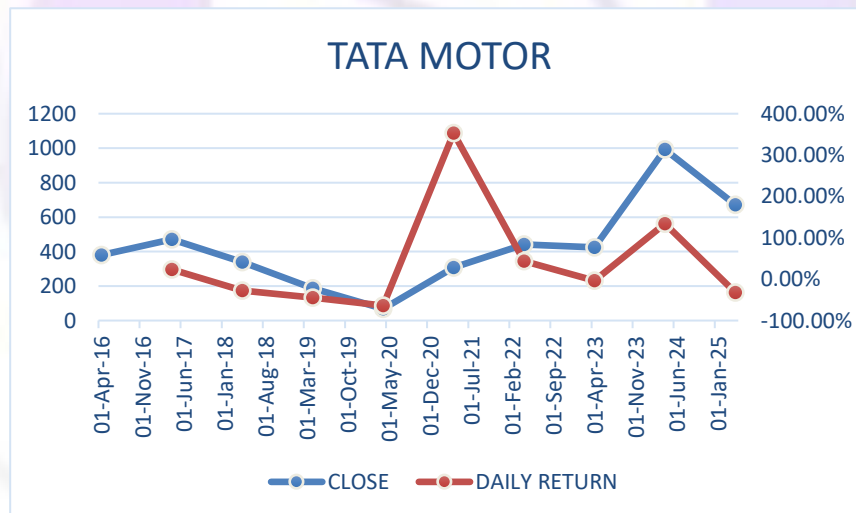
Date	TATA				
	OPEN	HIGH	LOW	CLOSE	DAILY RETURN
03-Apr-23	423	424.95	419.5	424.25	
02-May-23	482.5	484	474.55	480.25	13.20%
01-Jun-23	527.5	536.25	522.85	535.2	11.44%
03-Jul-23	600	602.3	590	590.8	10.39%
01-Aug-23	645	652.9	642	643.65	8.95%
01-Sep-23	604.2	614.9	601.9	611.2	-5.04%
03-Oct-23	632.6	632.65	618.6	620.45	1.51%
01-Nov-23	630	633.8	626.3	627.7	1.17%
01-Dec-23	708	711	703	705.45	12.39%
01-Jan-24	785	798.7	781.05	790.6	12.07%
01-Feb-24	900	900.15	876.3	878.5	11.12%
01-Mar-24	958.95	980.4	956.7	977.4	11.26%
01-Apr-24	1,000.00	1,009.85	987.45	992.25	1.52%
02-May-24	1,020.00	1,030.00	1,011.55	1,027.80	3.58%
03-Jun-24	963	963	934	950.45	-7.53%
01-Jul-24	989.75	1,005.50	985.2	1,002.05	5.43%
01-Aug-24	1,167.00	1,176.00	1,136.50	1,144.40	14.21%
02-Sep-24	1,105.00	1,105.00	1,087.35	1,092.65	-4.52%
01-Oct-24	976.9	984.5	958.9	965.2	-11.66%
01-Nov-24	847.95	847.95	840	843.45	-12.61%
02-Dec-24	787.3	796.75	785	790.05	-6.33%
01-Jan-25	740.45	751.1	736	749.25	-5.16%
01-Feb-25	719.5	724.75	700.9	706.55	-5.70%
03-Mar-25	620.65	626.85	606.3	621.15	-12.09%



The monthly return trend of Tata Motors reflects pronounced volatility, with sharp upward movements followed by noticeable declines. Such swings indicate the stock's high sensitivity to market dynamics and sectoral developments. While it offers periods of strong gains, the irregular pattern highlights that investors face elevated short-term risks compared to more stable peers.

Yearly Stock Price of TATA Motors

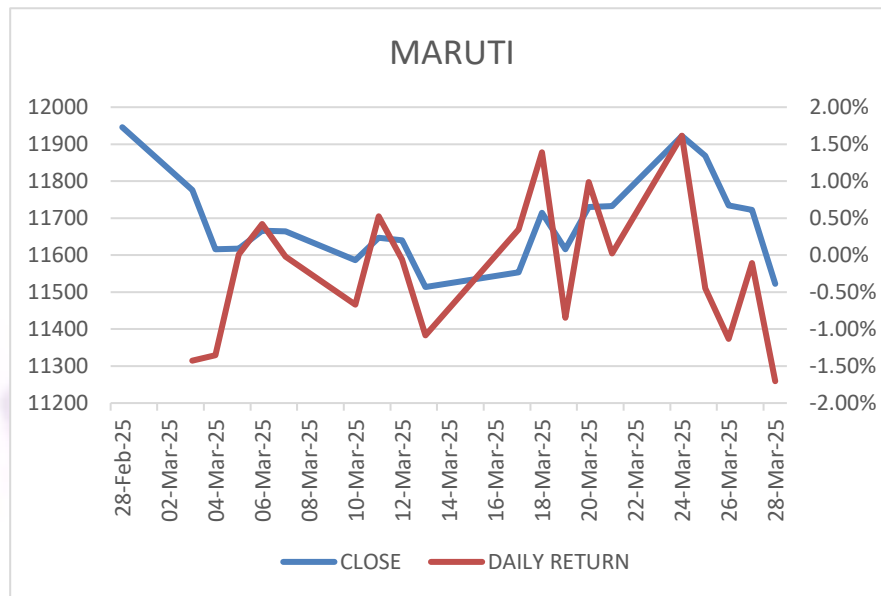
Date	TATA				
	OPEN	HIGH	LOW	CLOSE	DAILY RETURN
01-Apr-16	386.4	386.4	377.7	379.65	
03-Apr-17	468.25	472.3	467.6	470.4	23.90%
02-Apr-18	335	340.8	332.7	339.15	-27.90%
01-Apr-19	176.35	189.75	176.35	187.35	-44.76%
01-Apr-20	70.5	71.65	67.55	67.95	-63.73%
01-Apr-21	306.75	309.85	303.05	307.75	352.91%
01-Apr-22	434	442.25	430.85	441.15	43.35%
03-Apr-23	423	424.95	419.5	424.25	-3.83%
01-Apr-24	1,000.00	1,009.85	987.45	992.25	133.88%
01-Apr-25	670.55	681.85	665.75	671.85	-32.29%



The yearly return trend of Tata Motors highlights pronounced volatility, with periods of robust gains followed by steep declines. Such cyclical swings underscore the company's exposure to global market conditions, currency fluctuations, and industry-specific challenges. While the stock offers considerable growth opportunities, it also carries a high degree of long-term risk for investors.

Daily Stock Price of Maruti Suzuki

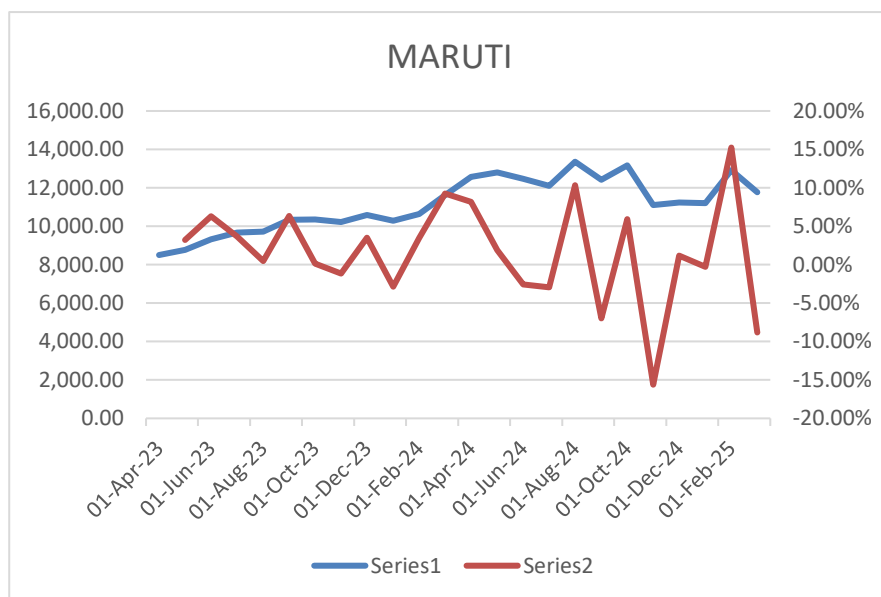
Date	Maruti				
	OPEN	HIGH	LOW	CLOSE	DAILY RETURN
28-Feb-25	12338.5	12338.5	11811.7	11945.85	
03-Mar-25	11886	12059.45	11757.75	11775.6	-1.43%
04-Mar-25	11690	11724.25	11591	11616.25	-1.35%
05-Mar-25	11605	11830	11568.35	11617.35	0.01%
06-Mar-25	11701	11729.5	11480	11666.4	0.42%
07-Mar-25	11650	11742.3	11601	11664.15	-0.02%
10-Mar-25	11600	11697.9	11533.65	11586.05	-0.67%
11-Mar-25	11579.5	11692	11510	11646.95	0.53%
12-Mar-25	11620.8	11723.95	11522	11640.1	-0.06%
13-Mar-25	11634.8	11685	11489.75	11513.8	-1.09%
17-Mar-25	11513.8	11752.95	11456.35	11553.7	0.35%
18-Mar-25	11560.05	11755.45	11501	11714.55	1.39%
19-Mar-25	11700	11714.45	11547	11615.55	-0.85%
20-Mar-25	11699.9	11765	11588	11730.2	0.99%
21-Mar-25	11780	11921.7	11705.8	11732.8	0.02%
24-Mar-25	11779.95	11990	11742.05	11922.25	1.61%
25-Mar-25	11951.4	12075.25	11800	11868.9	-0.45%
26-Mar-25	11807	11868.9	11710	11734.3	-1.13%
27-Mar-25	11773.15	11898	11638	11721.95	-0.11%
28-Mar-25	11699.05	11745.95	11400.1	11522.15	-1.70%



The daily return chart for Maruti Suzuki shows a stable trend with mostly negative returns. The stock exhibits limited fluctuations, indicating consistent behavior. Gains and losses are relatively small, and the chart reflects a conservative movement. Overall, Maruti Suzuki's daily performance is subdued.

Monthly Stock Price of Maruti Suzuki

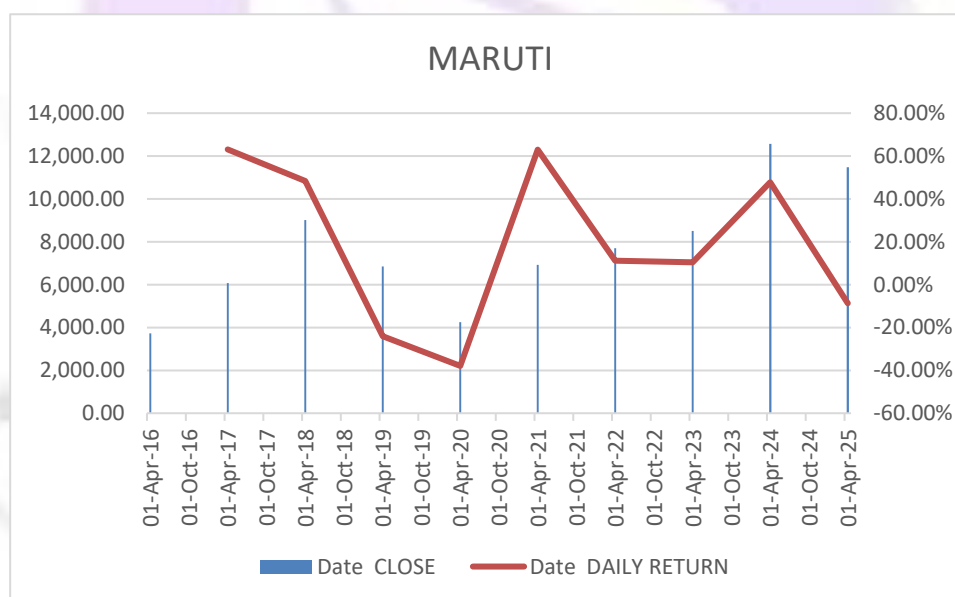
Date	Maruti				
	OPEN	HIGH	LOW	CLOSE	DAILY RETURN
03-Apr-23	8,412.85	8,579.00	8,390.35	8,503.05	
02-May-23	8,622.70	8,827.00	8,606.00	8,776.85	3.22%
01-Jun-23	9,337.50	9,377.55	9,222.00	9,328.30	6.28%
03-Jul-23	9,742.05	9,785.85	9,656.55	9,673.00	3.70%
01-Aug-23	9,760.00	9,892.00	9,650.05	9,716.75	0.45%
01-Sep-23	9,999.00	10,397.95	9,951.35	10,331.25	6.32%
03-Oct-23	10,709.80	10,709.80	10,303.30	10,346.90	0.15%
01-Nov-23	10,390.10	10,422.20	10,188.65	10,227.75	-1.15%
01-Dec-23	10,608.00	10,759.00	10,533.05	10,585.70	3.50%
01-Jan-24	10,345.00	10,345.00	10,214.65	10,283.30	-2.86%
01-Feb-24	10,210.00	10,699.80	10,204.00	10,637.95	3.45%
01-Mar-24	11,295.00	11,675.00	11,295.00	11,620.95	9.24%
01-Apr-24	12,613.35	12,689.00	12,506.15	12,569.60	8.16%
02-May-24	12,750.00	12,844.15	12,514.80	12,805.35	1.88%
03-Jun-24	12,825.00	12,825.00	12,402.55	12,476.15	-2.57%
01-Jul-24	12,045.00	12,260.00	12,005.55	12,108.65	-2.95%
01-Aug-24	13,393.00	13,680.00	13,112.00	13,359.05	10.33%
02-Sep-24	12,373.00	12,525.00	12,350.40	12,427.40	-6.97%
01-Oct-24	13,238.00	13,300.45	13,000.05	13,166.00	5.94%
01-Nov-24	11,150.00	11,209.55	11,080.00	11,110.00	-15.62%
02-Dec-24	11,151.00	11,330.00	11,151.00	11,239.30	1.16%
01-Jan-25	10,858.00	11,273.95	10,800.20	11,208.30	-0.28%
01-Feb-25	12,320.00	13,150.00	12,228.65	12,916.50	15.24%
03-Mar-25	11,886.00	12,059.45	11,757.75	11,775.60	-8.83%



Maruti's monthly return pattern is marked by controlled and steady performance, with fewer extreme variations. The stock demonstrates resilience and defensive qualities, as returns tend to cluster around stable, moderate gains. This stability makes Maruti a lower-risk option for investors seeking reliability over aggressive short-term returns.

Yearly Stock Price of Maruti Suzuki

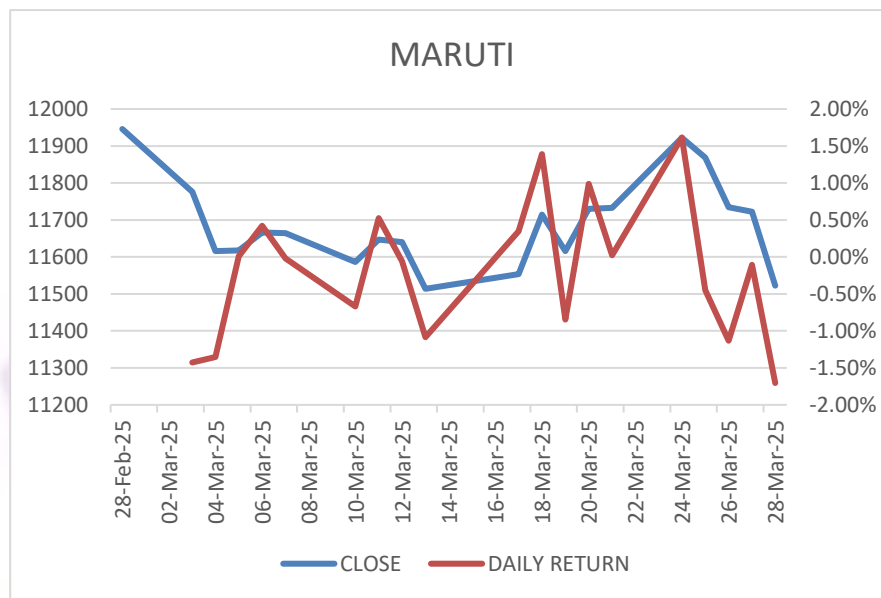
Date	Maruti				
	OPEN	HIGH	LOW	CLOSE	DAILY RETURN
01-Apr-16	3,700.45	3,764.95	3,682.00	3,723.20	
03-Apr-17	6,095.00	6,095.00	6,030.00	6,072.00	63.09%
02-Apr-18	8,990.00	9,088.00	8,953.25	9,008.20	48.36%
01-Apr-19	6,730.80	6,870.00	6,700.00	6,840.70	-24.06%
01-Apr-20	4,290.00	4,424.85	4,190.10	4,246.35	-37.93%
01-Apr-21	6,950.00	6,991.50	6,855.00	6,923.90	63.06%
01-Apr-22	7,574.00	7,729.00	7,517.35	7,700.05	11.21%
03-Apr-23	8,412.85	8,579.00	8,390.35	8,503.05	10.43%
01-Apr-24	12,613.35	12,689.00	12,506.15	12,569.60	47.82%
01-Apr-25	11,480.00	11,545.80	11,290.00	11,481.10	-8.66%



Maruti's yearly performance exhibits remarkable consistency, with steady gains and only limited downside phases. This stability reflects its dominant market position and efficient business model, which help it withstand sectoral pressures. Maruti stands out as a defensive and dependable choice, offering investors sustained long-term value with lower risk exposure.

Daily Stock Price of Mahindra & Mahindra Ltd

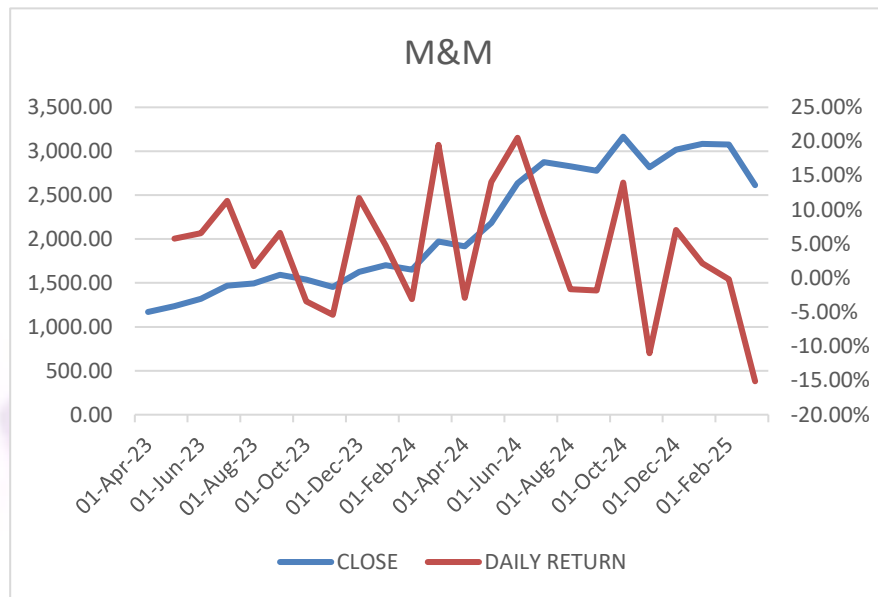
Date	M&M				
	OPEN	HIGH	LOW	CLOSE	DAILY RETURN
28-Feb-25	2695.1	2699	2572.2	2585.1	
03-Mar-25	2613.8	2695.95	2595	2611.9	1.04%
04-Mar-25	2561	2627.3	2550	2613.35	0.06%
05-Mar-25	2661.95	2736.9	2642.65	2726.6	4.33%
06-Mar-25	2748	2766.85	2695	2742.4	0.58%
07-Mar-25	2721.3	2765.95	2708.1	2727.85	-0.53%
10-Mar-25	2695	2734.65	2665.1	2702.6	-0.93%
11-Mar-25	2684.75	2685	2613.15	2645.6	-2.11%
12-Mar-25	2645.6	2678	2636.1	2653.3	0.29%
13-Mar-25	2677.1	2686.35	2618.3	2643.5	-0.37%
17-Mar-25	2635	2727.05	2635	2705.15	2.33%
18-Mar-25	2732.3	2798	2715.05	2791.4	3.19%
19-Mar-25	2795	2831.9	2779.55	2789.1	-0.08%
20-Mar-25	2803.75	2847	2801	2828.1	1.40%
21-Mar-25	2839.1	2893	2781.25	2801.85	-0.93%
24-Mar-25	2815.3	2822.15	2709.4	2774.7	-0.97%
25-Mar-25	2774.7	2802.45	2725.05	2736	-1.39%
26-Mar-25	2751.3	2795	2737.05	2742.25	0.23%
27-Mar-25	2735	2751	2695.55	2733	-0.34%
28-Mar-25	2735.8	2739.7	2635	2665.8	-2.46%



The daily return chart for M&M shows a fluctuating trend with mostly positive returns. The stock experiences noticeable ups and downs, reflecting short-term movements. The chart captures both gains and losses across trading days. Overall, M&M's stock shows dynamic behavior in the short term.

Monthly Stock Price of Mahindra & Mahindra Ltd

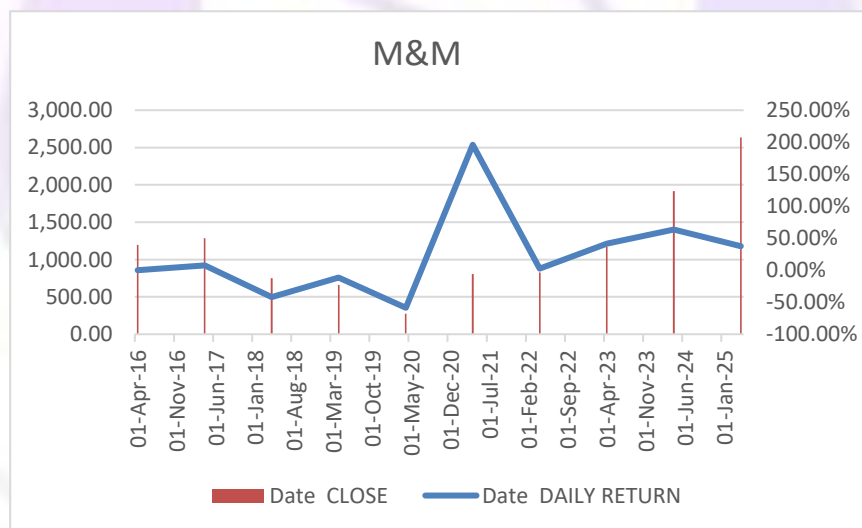
Date	M&M				
	OPEN	HIGH	LOW	CLOSE	DAILY RETURN
03-Apr-23	1,161.30	1,180.00	1,155.30	1,169.90	
02-May-23	1,230.00	1,240.80	1,217.50	1,237.30	5.76%
01-Jun-23	1,319.00	1,337.70	1,312.80	1,318.90	6.60%
03-Jul-23	1,470.00	1,498.05	1,459.30	1,467.95	11.30%
01-Aug-23	1,475.20	1,504.00	1,470.20	1,493.35	1.73%
01-Sep-23	1,577.00	1,604.50	1,575.00	1,591.95	6.60%
03-Oct-23	1,597.20	1,598.00	1,533.05	1,537.40	-3.43%
01-Nov-23	1,464.90	1,485.00	1,450.00	1,454.60	-5.39%
01-Dec-23	1,656.00	1,666.00	1,621.20	1,625.50	11.75%
01-Jan-24	1,729.40	1,730.00	1,699.00	1,703.30	4.79%
01-Feb-24	1,659.25	1,702.60	1,645.55	1,651.15	-3.06%
01-Mar-24	1,933.60	1,982.30	1,930.50	1,972.95	19.49%
01-Apr-24	1,938.00	1,963.85	1,891.80	1,915.95	-2.89%
02-May-24	2,165.10	2,204.00	2,165.10	2,185.15	14.05%
03-Jun-24	2,575.05	2,654.35	2,570.40	2,634.00	20.54%
01-Jul-24	2,841.00	2,894.05	2,838.25	2,875.85	9.18%
01-Aug-24	2,926.50	2,926.50	2,820.10	2,828.40	-1.65%
02-Sep-24	2,848.80	2,848.80	2,756.50	2,777.00	-1.82%
01-Oct-24	3,105.50	3,185.45	3,095.80	3,165.50	13.99%
01-Nov-24	2,780.00	2,840.90	2,780.00	2,817.65	-10.99%
02-Dec-24	2,984.00	3,026.25	2,961.90	3,016.40	7.05%
01-Jan-25	3,007.10	3,086.95	2,995.20	3,082.00	2.17%
01-Feb-25	2,990.00	3,098.95	2,990.00	3,076.75	-0.17%
03-Mar-25	2,613.80	2,695.95	2,595.00	2,611.90	-15.11%



M&M's monthly returns display a healthy balance of growth and risk, with consistent upward momentum interspersed with limited corrections. The presence of multiple positive months suggests strong investor confidence and favorable sectoral performance. Compared to Tata Motors, M&M offers a steadier growth profile with moderate volatility, making it attractive for balanced investors.

Yearly Stock Price of Mahindra & Mahindra Ltd

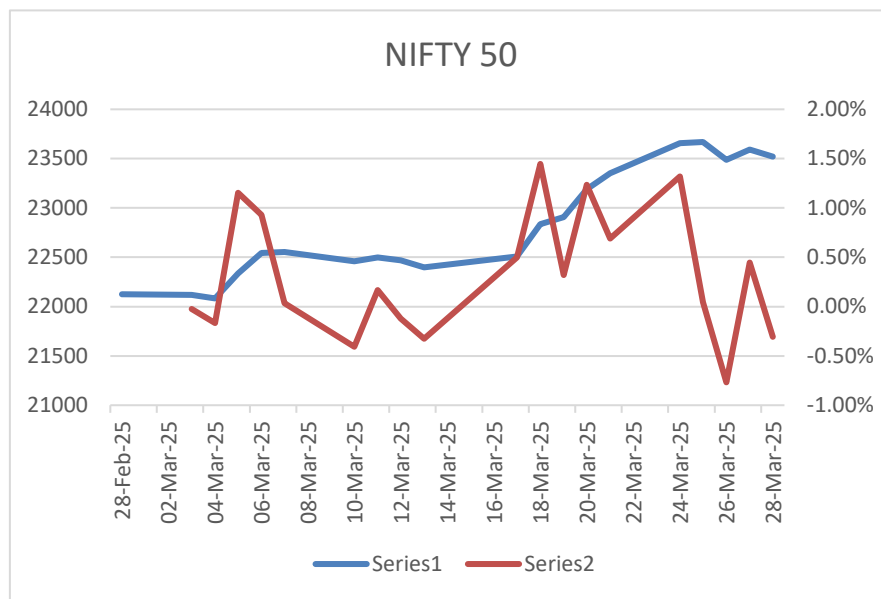
Date	M&M				
	OPEN	HIGH	LOW	CLOSE	DAILY RETURN
01-Apr-16	1,206.70	1,207.40	1,188.00	1,195.20	
03-Apr-17	1,297.60	1,305.00	1,280.00	1,286.50	7.64%
02-Apr-18	748.9	751.85	737.2	748.05	-41.85%
01-Apr-19	678.15	689	659	660.9	-11.65%
01-Apr-20	281	283	265.5	272.85	-58.72%
01-Apr-21	801.5	809.8	779.75	807.6	195.99%
01-Apr-22	811	829.5	807.7	826.55	2.35%
03-Apr-23	1,161.30	1,180.00	1,155.30	1,169.90	41.54%
01-Apr-24	1,938.00	1,963.85	1,891.80	1,915.95	63.77%
01-Apr-25	2,652.80	2,728.00	2,623.15	2,637.90	37.68%



M&M's yearly returns reflect a healthier balance of growth and correction, with several years of strong appreciation punctuated by manageable downturns. The company benefits from its diverse portfolio across automobiles and agriculture, which cushions volatility. Overall, M&M demonstrates resilience and consistent investor confidence, making it a relatively stable long-term performer.

Daily Stock Price of NSE Index

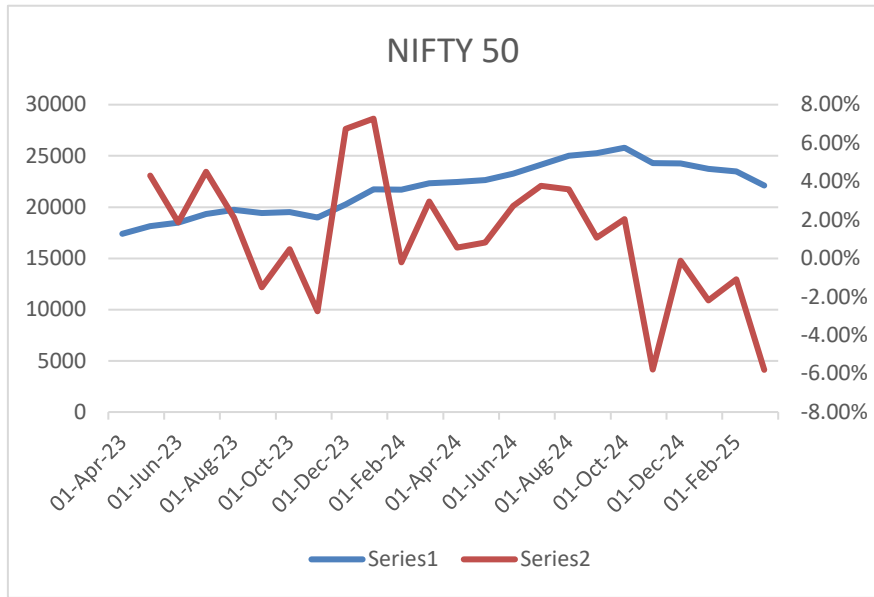
Date	NIFTY 50				
	Open	High	Low	Close	DAILY RETURN
28-Feb-25	22433.4	22450.35	22104.85	22124.7	
03-Mar-25	22194.55	22261.55	22004.7	22119.3	-0.02%
04-Mar-25	21974.45	22105.05	21964.6	22082.65	-0.17%
05-Mar-25	22073.05	22394.9	22067.8	22337.3	1.15%
06-Mar-25	22476.35	22556.45	22245.85	22544.7	0.93%
07-Mar-25	22508.65	22633.8	22464.75	22552.5	0.03%
10-Mar-25	22521.85	22676.75	22429.05	22460.3	-0.41%
11-Mar-25	22345.95	22522.1	22314.7	22497.9	0.17%
12-Mar-25	22536.35	22577.4	22329.55	22470.5	-0.12%
13-Mar-25	22541.5	22558.05	22377.35	22397.2	-0.33%
17-Mar-25	22353.15	22577	22353.15	22508.75	0.50%
18-Mar-25	22662.25	22857.8	22599.2	22834.3	1.45%
19-Mar-25	22874.95	22940.7	22807.95	22907.6	0.32%
20-Mar-25	23036.6	23216.7	22973.95	23190.65	1.24%
21-Mar-25	23168.25	23402.7	23132.8	23350.4	0.69%
24-Mar-25	23515.4	23708.75	23433.5	23658.35	1.32%
25-Mar-25	23751.5	23869.6	23601.4	23668.65	0.04%
26-Mar-25	23700.95	23736.5	23451.7	23486.85	-0.77%
27-Mar-25	23433.95	23646.45	23412.2	23591.95	0.45%
28-Mar-25	23600.4	23649.2	23450.2	23519.35	-0.31%



The daily return chart for Nifty 50 shows a stable trend with mostly positive returns. The index moves steadily with minor fluctuations, reflecting overall market stability. Peaks and dips are mild, indicating consistent performance. The chart portrays Nifty 50 as a reliable benchmark.

Monthly Stock Price of NSE Index

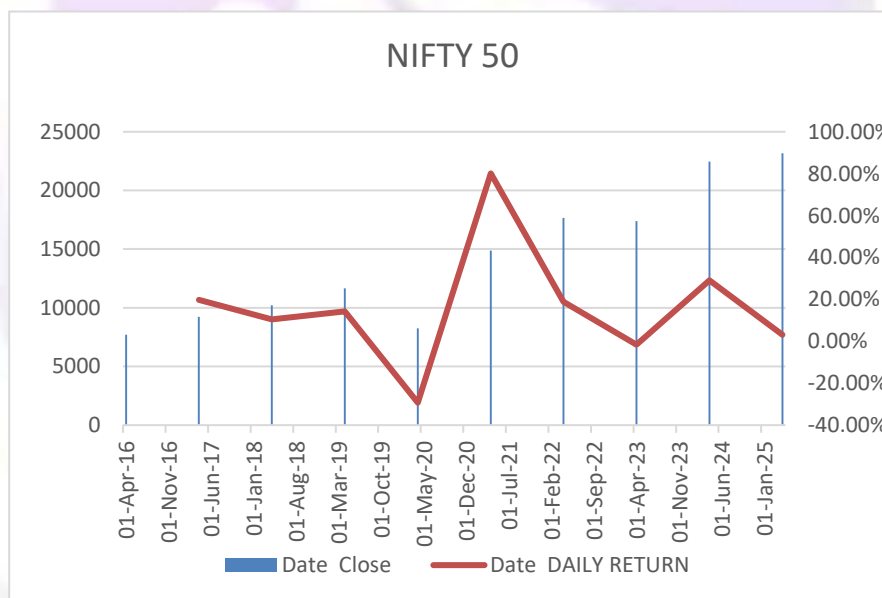
Date	NIFTY 50				
	Open	High	Low	Close	DAILY RETURN
03-Apr-23	17427.95	17428.05	17312.75	17398.05	
02-May-23	18124.8	18180.25	18101.75	18147.65	4.31%
01-Jun-23	18579.4	18580.3	18464.55	18487.75	1.87%
03-Jul-23	19246.5	19345.1	19234.4	19322.55	4.52%
01-Aug-23	19784	19795.6	19704.6	19733.55	2.13%
01-Sep-23	19258.15	19458.55	19255.7	19435.3	-1.51%
03-Oct-23	19622.4	19623.2	19479.65	19528.75	0.48%
01-Nov-23	19064.05	19096.05	18973.7	18989.15	-2.76%
01-Dec-23	20194.1	20291.55	20183.7	20267.9	6.73%
01-Jan-24	21727.75	21834.35	21680.85	21741.9	7.27%
01-Feb-24	21780.65	21832.95	21658.75	21697.45	-0.20%
01-Mar-24	22048.3	22353.3	22047.75	22338.75	2.96%
01-Apr-24	22455	22529.95	22427.75	22462	0.55%
02-May-24	22567.85	22710.5	22567.85	22648.2	0.83%
03-Jun-24	23337.9	23338.7	23062.3	23263.9	2.72%
01-Jul-24	23992.95	24164	23992.7	24141.95	3.77%
01-Aug-24	25030.95	25078.3	24956.4	25010.9	3.60%
02-Sep-24	25333.6	25333.65	25235.5	25278.7	1.07%
01-Oct-24	25788.45	25907.6	25739.2	25796.9	2.05%
01-Nov-24	24302.75	24368.25	24280.2	24304.35	-5.79%
02-Dec-24	24140.85	24301.7	24008.65	24276.05	-0.12%
01-Jan-25	23637.65	23822.8	23562.8	23742.9	-2.20%
01-Feb-25	23528.6	23632.45	23318.3	23482.15	-1.10%
03-Mar-25	22194.55	22261.55	22004.7	22119.3	-5.80%



The NIFTY 50 monthly returns are relatively smooth and stable, with minor positive and negative changes spread evenly across months. The absence of extreme spikes reinforces the diversification advantage of the index, which dampens volatility. Serving as a reliable benchmark, NIFTY highlights how market-wide risks are more contained than those of individual stocks.

Yearly Stock Price of NSE Index

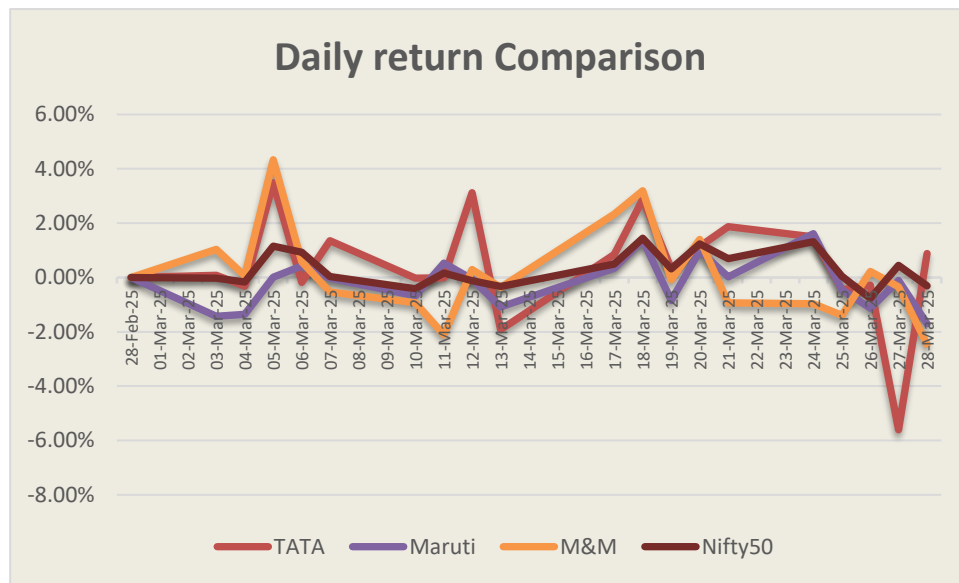
Date	NIFTY 50				
	Open	High	Low	Close	DAILY RETURN
01-Apr-16	7718.05	7740.15	7666.1	7713.05	
03-Apr-17	9220.6	9245.35	9192.4	9237.85	19.77%
02-Apr-18	10151.65	10220.1	10127.75	10211.8	10.54%
01-Apr-19	11665.2	11738.1	11644.75	11669.15	14.27%
01-Apr-20	8584.1	8588.1	8198.35	8253.8	-29.27%
01-Apr-21	14798.4	14883.2	14692.45	14867.35	80.13%
01-Apr-22	17436.9	17703.7	17422.7	17670.45	18.85%
03-Apr-23	17427.95	17428.05	17312.75	17398.05	-1.54%
01-Apr-24	22455	22529.95	22427.75	22462	29.11%
01-Apr-25	23,341.10	23,565.15	23,136.40	23,165.70	3.13%



The yearly returns of NIFTY 50 remain measured and stable, showing moderate growth without extreme fluctuations. As a diversified index, it smooths out sector-specific shocks, thereby reducing volatility for long-term investors. NIFTY thus serves as a trusted benchmark, demonstrating how diversification consistently delivers predictable and less risky returns compared to individual stocks.

Daily return comparison of the companies

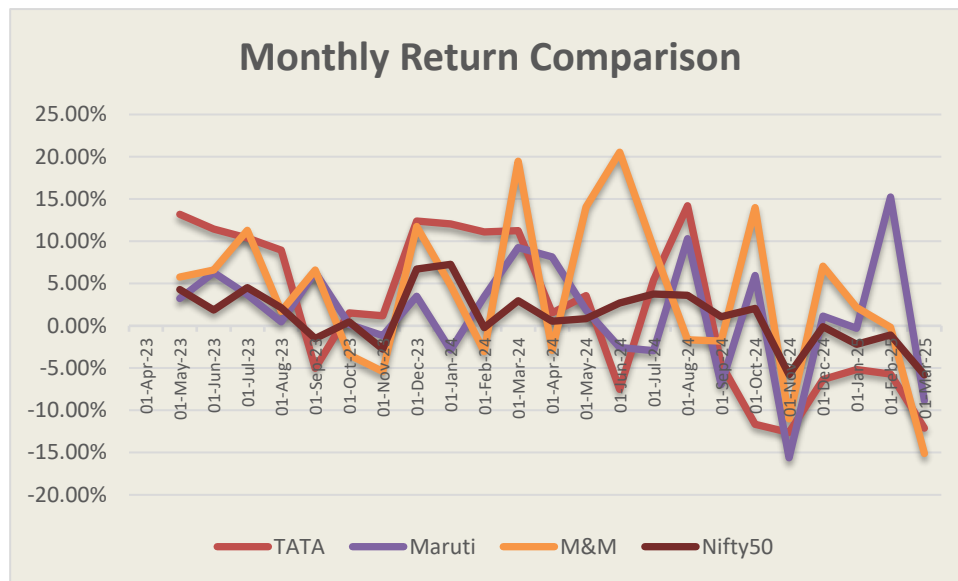
Date	TATA		Maruti		M&M		Nifty50	
	CLOSE	DAILY RETURN	CLOSE	DAILY RETURN	CLOSE	DAILY RETURN	Close	DAILY RETURN
28-Feb-25	620.65		11945.85		2585.1		22124.7	
03-Mar-25	621.15	0.08%	11775.6	-1.43%	2611.9	1.04%	22119.3	-0.02%
04-Mar-25	619.2	-0.31%	11616.25	-1.35%	2613.35	0.06%	22082.65	-0.17%
05-Mar-25	640.8	3.49%	11617.35	0.01%	2726.6	4.33%	22337.3	1.15%
06-Mar-25	639.65	-0.18%	11666.4	0.42%	2742.4	0.58%	22544.7	0.93%
07-Mar-25	648.3	1.35%	11664.15	-0.02%	2727.85	-0.53%	22552.5	0.03%
10-Mar-25	648.15	-0.02%	11586.05	-0.67%	2702.6	-0.93%	22460.3	-0.41%
11-Mar-25	648.05	-0.02%	11646.95	0.53%	2645.6	-2.11%	22497.9	0.17%
12-Mar-25	668.3	3.12%	11640.1	-0.06%	2653.3	0.29%	22470.5	-0.12%
13-Mar-25	655.5	-1.92%	11513.8	-1.09%	2643.5	-0.37%	22397.2	-0.33%
17-Mar-25	661.05	0.85%	11553.7	0.35%	2705.15	2.33%	22508.75	0.50%
18-Mar-25	680.05	2.87%	11714.55	1.39%	2791.4	3.19%	22834.3	1.45%
19-Mar-25	682.25	0.32%	11615.55	-0.85%	2789.1	-0.08%	22907.6	0.32%
20-Mar-25	690.05	1.14%	11730.2	0.99%	2828.1	1.40%	23190.65	1.24%
21-Mar-25	702.95	1.87%	11732.8	0.02%	2801.85	-0.93%	23350.4	0.69%
24-Mar-25	713.5	1.50%	11922.25	1.61%	2774.7	-0.97%	23658.35	1.32%
25-Mar-25	710.3	-0.45%	11868.9	-0.45%	2736	-1.39%	23668.65	0.04%
26-Mar-25	708.25	-0.29%	11734.3	-1.13%	2742.25	0.23%	23486.85	-0.77%
27-Mar-25	668.55	-5.61%	11721.95	-0.11%	2733	-0.34%	23591.95	0.45%
28-Mar-25	674.45	0.88%	11522.15	-1.70%	2665.8	-2.46%	23519.35	-0.31%



This chart shows the daily returns and closing prices for Tata Motors, Maruti, M&M, and NIFTY 50 for a recent period (e.g., days or weeks). It visualizes the day-to-day price volatility and relative return fluctuations for the companies and the market index. Trends of peaks and troughs indicate how volatile and responsive each stock is relative to the index.

Monthly return comparison of the companies

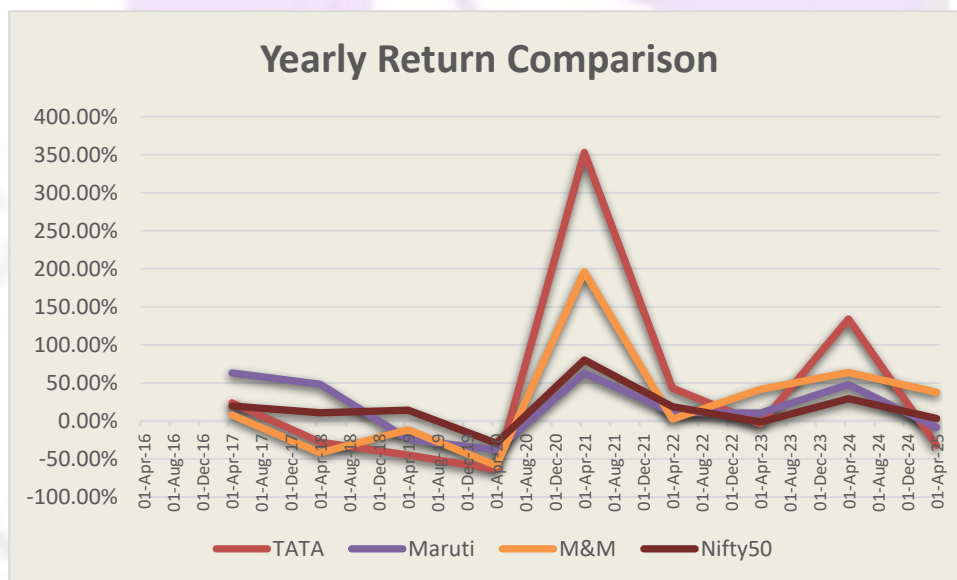
Date	TATA		Maruti		M&M		Nifty50	
	CLOSE	DAILY RETURN	CLOSE	DAILY RETURN	CLOSE	DAILY RETURN	Close	DAILY RETURN
03-Apr-23	424.25		8,503.05		1,169.90		17398.05	
02-May-23	480.25	13.20%	8,776.85	3.22%	1,237.30	5.76%	18147.65	4.31%
01-Jun-23	535.2	11.44%	9,328.30	6.28%	1,318.90	6.60%	18487.75	1.87%
03-Jul-23	590.8	10.39%	9,673.00	3.70%	1,467.95	11.30%	19322.55	4.52%
01-Aug-23	643.65	8.95%	9,716.75	0.45%	1,493.35	1.73%	19733.55	2.13%
01-Sep-23	611.2	-5.04%	10,331.25	6.32%	1,591.95	6.60%	19435.3	-1.51%
03-Oct-23	620.45	1.51%	10,346.90	0.15%	1,537.40	-3.43%	19528.75	0.48%
01-Nov-23	627.7	1.17%	10,227.75	-1.15%	1,454.60	-5.39%	18989.15	-2.76%
01-Dec-23	705.45	12.39%	10,585.70	3.50%	1,625.50	11.75%	20267.9	6.73%
01-Jan-24	790.6	12.07%	10,283.30	-2.86%	1,703.30	4.79%	21741.9	7.27%
01-Feb-24	878.5	11.12%	10,637.95	3.45%	1,651.15	-3.06%	21697.45	-0.20%
01-Mar-24	977.4	11.26%	11,620.95	9.24%	1,972.95	19.49%	22338.75	2.96%
01-Apr-24	992.25	1.52%	12,569.60	8.16%	1,915.95	-2.89%	22462	0.55%
02-May-24	1,027.80	3.58%	12,805.35	1.88%	2,185.15	14.05%	22648.2	0.83%
03-Jun-24	950.45	-7.53%	12,476.15	-2.57%	2,634.00	20.54%	23263.9	2.72%
01-Jul-24	1,002.05	5.43%	12,108.65	-2.95%	2,875.85	9.18%	24141.95	3.77%
01-Aug-24	1,144.40	14.21%	13,359.05	10.33%	2,828.40	-1.65%	25010.9	3.60%
02-Sep-24	1,092.65	-4.52%	12,427.40	-6.97%	2,777.00	-1.82%	25278.7	1.07%
01-Oct-24	965.2	-11.66%	13,166.00	5.94%	3,165.50	13.99%	25796.9	2.05%
01-Nov-24	843.45	-12.61%	11,110.00	-15.62%	2,817.65	-10.99%	24304.35	-5.79%
02-Dec-24	790.05	-6.33%	11,239.30	1.16%	3,016.40	7.05%	24276.05	-0.12%
01-Jan-25	749.25	-5.16%	11,208.30	-0.28%	3,082.00	2.17%	23742.9	-2.20%
01-Feb-25	706.55	-5.70%	12,916.50	15.24%	3,076.75	-0.17%	23482.15	-1.10%
03-Mar-25	621.15	-12.09%	11,775.60	-8.83%	2,611.90	-15.11%	22119.3	-5.80%



This chart compares the monthly returns of Tata Motors, Maruti, M&M, and NIFTY 50 over a significant timeframe, covering multiple months. It highlights which months showed positive or negative performance for each stock versus the overall index. This aids in understanding medium-term performance and seasonality or market cycle impact on these stocks.

Annually return comparison of the companies

Date	TATA		Maruti		M&M		Nifty50	
	CLOSE	DAILY RETURN	CLOSE	DAILY RETURN	CLOSE	DAILY RETURN	Close	DAILY RETURN
01-Apr-16	379.65		3,723.20		1,195.20		7713.05	
03-Apr-17	470.4	23.90%	6,072.00	63.09%	1,286.50	7.64%	9237.85	19.77%
02-Apr-18	339.15	-27.90%	9,008.20	48.36%	748.05	-41.85%	10211.8	10.54%
01-Apr-19	187.35	-44.76%	6,840.70	-24.06%	660.9	-11.65%	11669.15	14.27%
01-Apr-20	67.95	-63.73%	4,246.35	-37.93%	272.85	-58.72%	8253.8	-29.27%
01-Apr-21	307.75	352.91%	6,923.90	63.06%	807.6	195.99%	14867.35	80.13%
01-Apr-22	441.15	43.35%	7,700.05	11.21%	826.55	2.35%	17670.45	18.85%
03-Apr-23	424.25	-3.83%	8,503.05	10.43%	1,169.90	41.54%	17398.05	-1.54%
01-Apr-24	992.25	133.88%	12,569.60	47.82%	1,915.95	63.77%	22462	29.11%
01-Apr-25	671.85	-32.29%	11,481.10	-8.66%	2,637.90	37.68%	23,165.70	3.13%



This chart presents yearly returns over several years for the same companies and the NIFTY 50. It shows long-term performance trends, growth sustainability, and relative out- or under-performance against the market index. It helps investors understand how the companies have fared over business cycles and macroeconomic conditions.

Daily descriptive statistics of the company:

TATA Motors		M&M		Maruti	
Mean	0.004577	Mean	0.001757052	Mean	-0.00186
Standard Error	0.004595	Standard Error	0.00393434	Standard Error	0.002166
Median	0.003235	Median	-0.000823959	Median	-0.00059
Mode	#N/A	Mode	#N/A	Mode	#N/A
Standard Deviation	0.020029	Standard Deviat	0.017149391	Standard Deviation	0.009442
Sample Variance	0.000401	Sample Variance	0.000294102	Sample Variance	8.92E-05
Kurtosis	3.949635	Kurtosis	0.843598579	Kurtosis	-0.61274
Skewness	-1.33199	Skewness	0.896800569	Skewness	0.231362
Range	0.090937	Range	0.067923547	Range	0.033192
Minimum	-0.05605	Minimum	-0.024588364	Minimum	-0.01704
Maximum	0.034884	Maximum	0.043335183	Maximum	0.016147
Sum	0.086967	Sum	0.033383982	Sum	-0.03528
Count	19	Count	19	Count	19
Smallest(1)	-0.05605	Smallest(1)	-0.024588364	Smallest(1)	-0.01704
Q1	-0.00234	Q1	-0.009269112	Q1	-0.00965
Q2	0.003235	Q2	-0.000823959	Q2	-0.00059
Q3	0.014266	Q3	0.008080933	Q3	0.003844

The descriptive analysis of daily returns for Tata Motors, M&M, Maruti, and the Nifty 50 provides valuable insights into their performance, risk, and distributional characteristics. Among the auto stocks, Tata Motors shows the highest mean return, but this is accompanied by the largest standard deviation, negative skewness, and high kurtosis, reflecting both strong upside potential and significant downside risk. M&M records modest positive mean returns with moderate volatility, positive skewness, and relatively low kurtosis, indicating a more balanced profile with occasional larger gains. Maruti, in contrast, presents a negative average return, suggesting underperformance, though its low standard deviation highlights stability with limited extreme fluctuations. The Nifty 50 benchmark demonstrates a positive mean return and the lowest volatility, with mild positive skewness, confirming its status as a relatively safe and stable investment vehicle. Overall, the descriptive analysis reveals that while the auto sector offers opportunities for higher returns compared to the benchmark, it comes at the cost of greater risk and variability, making stock selection and investment horizon critical considerations for investors.

Monthly descriptive statistics of the company

TATA Motors		M&M		Maruti	
Mean	0.02068635	Mean	0.03935153	Mean	0.016437
Standard Error	0.01905485	Standard Error	0.01887657	Standard Error	0.014003
Median	0.01519337	Median	0.0478622	Median	0.018756
Mode	#N/A	Mode	#N/A	Mode	#N/A
Standard Deviation	0.09138386	Standard Deviation	0.09052883	Standard Deviation	0.067157
Sample Variance	0.00835101	Sample Variance	0.00819547	Sample Variance	0.00451
Kurtosis	-1.4562797	Kurtosis	-0.2421316	Kurtosis	1.06242
Skewness	-0.1964981	Skewness	-0.0710999	Skewness	-0.49928
Range	0.26819844	Range	0.35649399	Range	0.308565
Minimum	-0.1261397	Minimum	-0.1510847	Minimum	-0.15616
Maximum	0.14205878	Maximum	0.20540924	Maximum	0.152405
Sum	0.47578599	Sum	0.90508526	Sum	0.378049
Count	23	Count	23	Count	23
Smallest(1)	-0.1261397	Smallest(1)	-0.1510847	Smallest(1)	-0.15616
Q1	-0.0543163	Q1	-0.0235318	Q1	-0.01861
Q2	0.01519337	Q2	0.0478622	Q2	0.018756
Q3	0.11187982	Q3	0.10241468	Q3	0.061132

The descriptive analysis of daily returns for Tata Motors, M&M, Maruti, and the Nifty 50 reveals key insights into their performance, volatility, and distribution patterns. M&M shows the highest mean return (0.039352), supported by the largest median return, suggesting stronger overall performance within the sample period. Tata Motors and Maruti also exhibit positive mean returns (0.020686 and 0.016437 respectively), though Maruti's distribution is marked by higher kurtosis and stronger negative skewness, indicating occasional downside risks despite generally stable performance. In contrast, the Nifty 50, while recording the lowest mean return (0.011036), demonstrates the lowest standard deviation (0.0337), making it the most stable and predictable asset among the group. Both Tata Motors and M&M display higher volatility (standard deviations around 0.09), with broader ranges of returns, highlighting their riskier nature. The skewness values across stocks are generally negative, indicating a greater probability of adverse returns compared to extreme positive outcomes, though the extent varies. Overall, the analysis underscores that while the auto stocks, particularly M&M, offer higher return potential, they do so at the expense of greater volatility and downside risk, whereas the Nifty 50 maintains its role as a relatively safer benchmark with more consistent returns.

Yearly descriptive statistics of the company:

M&M		Maruti		Nifty 50	
Mean	0.263049109	Mean	0.192571	Mean	0.161104757
Standard Error	0.24942777	Standard Error	0.126768	Standard Error	0.097621591
Median	0.076388889	Median	0.112097	Median	0.142712352
Mode	#N/A	Mode	#N/A	Mode	#N/A
Standard Deviation	0.74828331	Standard Deviation	0.380303	Standard Deviation	0.292864774
Sample Variance	0.559927911	Sample Variance	0.144631	Sample Variance	0.085769776
Kurtosis	3.180790742	Kurtosis	-1.55155	Kurtosis	3.144279304
Skewness	1.510129007	Skewness	-0.21933	Skewness	1.044854288
Range	2.54702194	Range	1.010107	Range	1.093955339
Minimum	-0.58715388	Minimum	-0.37925	Minimum	-0.292681986
Maximum	1.959868059	Maximum	0.630855	Maximum	0.801273353
Sum	2.367441979	Sum	1.733137	Sum	1.44994281
Count	9	Count	9	Count	9
Smallest(1)	-0.58715388	Smallest(1)	-0.37925	Smallest(1)	-0.292681986
Confidence Level	0.575181469	Confidence Level	0.292327	Confidence Level	0.225115793
Q1	-0.11650291	Q1	-0.0866	Q1	0.031328466
Q2	0.076388889	Q2	0.112097	Q2	0.142712352
Q3	0.415401367	Q3	0.483564	Q3	0.197690926

The table provides a comparative descriptive analysis of the returns for TATA Motors, M&M, Maruti, and Nifty 50. TATA Motors shows the highest average return (0.4232) but also the highest volatility and risk, with a large standard deviation (1.3069), strong positive skewness (2.01), and high kurtosis (4.25), indicating extreme values. M&M has moderate returns (0.2630) and risk, with less skewness and lower variability. Maruti has lower returns (0.1926) and relatively low volatility but is negatively skewed (-2.19), suggesting potential for larger negative returns. Nifty 50 shows the lowest average return (0.1611) but is the most stable, with the smallest standard deviation (0.2983) and narrowest confidence interval, making it the least risky investment. Overall, TATA Motors suits high-risk investors, while Nifty 50 is better for conservative ones.

Correlation

Daily Correlation Analysis: Automobile Companies vs Nifty 50

Particular	Daily Correlation
Nifty50 and TATA Motors	0.887
Nifty50 and Maruti	0.286
Nifty50 and Mahindra & Mahindra	0.629

1. TATA Motors and Nifty 50

The relationship between TATA Motors and the Nifty 50 demonstrates a strong positive daily correlation of 0.887, indicating that the stock's price movements closely follow overall market trends. On most trading days, TATA Motors tends to move in the same direction as the Nifty 50, reflecting a high degree of alignment with broader market conditions. This suggests that the stock's performance is largely influenced by market-wide factors such as economic news, investor sentiment, and overall equity market movements. Due to this strong co-movement, TATA Motors offers limited diversification benefits when included in a portfolio that already has exposure to the Nifty 50.

2. Maruti Suzuki and Nifty 50

Maruti Suzuki exhibits a weak positive daily correlation of 0.286 with the Nifty 50, indicating a relatively low level of co-movement with the broader market. This implies that Maruti's stock price movements are less dependent on daily market fluctuations and are more influenced by company-specific factors, such as sales performance, demand trends, and operational efficiency. Because of this low correlation, Maruti Suzuki's returns do not closely mirror those of the index on a daily basis. From a portfolio perspective, this characteristic allows Maruti Suzuki to enhance diversification and reduce overall risk.

3. Mahindra & Mahindra and Nifty 50

Mahindra & Mahindra shows a moderate positive daily correlation of 0.629 with the Nifty 50, indicating partial alignment with market movements. While the stock generally follows the direction of the broader market, the relationship is not very strong, suggesting that firm-specific drivers such as rural demand, product mix, and segment performance also play a significant role in determining its price movements. As a result, Mahindra & Mahindra carries moderate market dependence and provides some diversification benefits, though less than Maruti Suzuki.

Monthly Correlation Analysis: Automobile Companies vs Nifty 50

Particular	Monthly Correlation
Nifty50 and TATA Motors	0.836
Nifty50 and Maruti	0.874
Nifty50 and Mahindra & Mahindra	0.930

1. TATA Motors and Nifty 50 (Monthly Correlation = 0.836)

TATA Motors shows a strong positive monthly correlation with the Nifty 50. This means that over a monthly time horizon, the stock generally moves in the same direction as the overall market. The strong relationship suggests that broader market trends play an important role in influencing TATA Motors' performance. Due to this high co-movement, TATA Motors offers limited diversification benefits when combined with market-index-based investments at the monthly level.

2. Maruti Suzuki and Nifty 50 (Monthly Correlation = 0.874)

Maruti Suzuki exhibits a strong positive correlation with the Nifty 50 on a monthly basis. This indicates that Maruti's returns tend to closely follow market movements over longer periods, even though it showed lower dependence in daily data. The strong monthly correlation suggests that market-wide economic conditions and investor sentiment influence Maruti more clearly over time. As a result, diversification benefits from Maruti reduce when the investment horizon shifts from daily to monthly.

3. Mahindra & Mahindra and Nifty 50 (Monthly Correlation = 0.930)

Mahindra & Mahindra demonstrates a very strong positive monthly correlation with the Nifty 50, indicating near-alignment with overall market performance. This suggests that over monthly intervals, M&M is highly sensitive to market trends and closely reflects index-level movements. Such a high correlation implies high exposure to systematic market risk and minimal diversification benefit when paired with the Nifty 50 over the monthly horizon.

Yearly Correlation Analysis: Automobile Companies vs Nifty 50

Particular	Yearly Correlation
Nifty50 and TATA Motors	0.756
Nifty50 and Maruti	0.874
Nifty50 and Mahindra & Mahindra	0.721

1. TATA Motors and Nifty 50 (Yearly Correlation = 0.756)

TATA Motors shows a strong positive correlation with the Nifty 50 on a yearly basis. This indicates that over longer time horizons, the stock generally moves in the same direction as the overall market, though not perfectly. The slightly lower correlation compared to monthly data suggests that company-specific factors such as business performance, strategy changes, and sector dynamics also influence returns over the year. Overall, TATA Motors remains market-linked but offers some scope for diversification in long-term portfolios.

2. Maruti Suzuki and Nifty 50 (Yearly Correlation = 0.874)

Maruti Suzuki exhibits a strong and consistent positive yearly correlation with the Nifty 50. This suggests that over annual periods, Maruti's performance is closely aligned with broader market trends. The high correlation indicates that macroeconomic conditions, market sentiment, and overall equity market movements play a significant role in shaping Maruti's annual returns. As a result, Maruti Suzuki provides limited diversification benefits when held alongside the index over a long-term horizon.

3. Mahindra & Mahindra and Nifty 50 (Yearly Correlation = 0.721)

Mahindra & Mahindra demonstrates a moderate to strong positive correlation with the Nifty 50 on a yearly basis. This suggests that while the stock generally follows market trends, it is not fully dependent on index movements. Factors such as rural demand, tractor sales, and segment-specific performance influence its returns over time. Because of this balanced relationship, Mahindra & Mahindra offers better diversification benefits compared to Maruti Suzuki and TATA Motors in a long-term portfolio.

Regression

Daily price

TATA Motors

Regression Statistics					
Multiple R		0.876530246			
R Square		0.768305273			
Adjusted R Square		0.754676171			
Standard Error		271.4471772			
Observations		19			
ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	4153720.254	4153720.254	56.37240776	8.56422E-07
Residual	17	1252620.69	73683.57001		
Total	18	5406340.944			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	11773.320	1476.805	7.972	0.000	8657.535	14889.105	8657.535	14889.105
620.65	16.599	2.211	7.508	0.000	11.935	21.264	11.935	21.264

RESIDUAL OUTPUT		
<i>Observation</i>	<i>Predicted 22124.7</i>	<i>Residuals</i>
1	22083.91133	35.38867123
2	22051.54289	31.10710606
3	22410.08556	-72.78555674
4	22390.99648	153.7035202
5	22534.57954	17.92046312
6	22532.08966	-71.78965728
7	22530.42974	-32.52973754
8	22866.56348	-396.0634839
9	22654.09376	-256.8937578
10	22746.2193	-237.4693031
11	23061.60405	-227.3040528
12	23098.12229	-190.522287
13	23227.59603	-36.94602632
14	23441.72567	-91.32567216
15	23616.8472	41.50279577
16	23563.72977	104.9202273
17	23529.70142	-42.85141813
18	22870.71328	721.2367167
19	22968.64855	550.7014524

Interpretation

The regression analysis between TATA Motors and the Nifty 50 indicates a strong and statistically significant linear relationship on a daily basis. The Multiple R value of 0.8765 reflects high co-movement, while the R^2 of 0.7683 shows that nearly 77% of the variation in the Nifty 50 is explained by movements in TATA Motors, confirming strong explanatory power. The Adjusted R^2 of 0.7547 closely aligns with R^2 , supporting the robustness of the model despite the limited sample size.

The slope coefficient of 16.60 is positive and highly significant ($p < 0.001$), indicating that a one-unit increase in TATA Motors' price leads to an average increase of approximately 16.6 units in the Nifty 50, highlighting the stock's strong market linkage. The F-statistic (56.37) further confirms overall model significance. Residuals are centered around zero, suggesting no systematic bias, though their spread indicates the presence of other market factors. Overall, the results confirm that TATA Motors is closely aligned with and significantly influences daily movements in the Nifty 50.

Maruti Suzuki

Regression Statistics	
Multiple R	0.521529863
R Square	0.271993398
Adjusted R Square	0.22916948
Standard Error	481.1659797
Observations	19

ANOVA					
	df	SS	MS	F	Significance F
Regression	1	1470489.043	1470489.043	6.35143658	0.022017152
Residual	17	3935851.901	231520.7001		
Total	18	5406340.944			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	-8269.377	12349.054	-0.670	0.512	34323.604	17784.851	34323.604	17784.851
11945.85	2.665	1.058	2.520	0.022	0.434	4.897	0.434	4.897

RESIDUAL OUTPUT		
Observation	Predicted 22124.7	Residuals
1	23117.14963	-997.8496302
2	22692.42028	-609.7702767
3	22695.3522	-358.052202
4	22826.08941	-281.3894143
5	22820.09229	-267.5922945
6	22611.9256	-151.6256017
7	22774.24765	-276.3476451
8	22755.98975	-285.489747
9	22419.35142	-22.15142041

10	22525.70035	-16.95034542
11	22954.42778	-120.1277787
12	22690.55451	217.0454939
13	22996.14108	194.5089211
14	23003.07108	347.328916
15	23508.02857	150.3214261
16	23365.8302	302.8198008
17	23007.06916	479.7808361
18	22974.15164	617.7983605
19	22441.6074	1077.742602

Interpretation

The daily regression analysis between Maruti Suzuki and the Nifty 50 indicates a moderate but statistically significant linear relationship. The Multiple R value of 0.5215 suggests a moderate degree of co-movement between Maruti's price and the index. The R^2 value of 0.2720 implies that approximately 27% of the variation in the Nifty 50 is explained by movements in Maruti Suzuki, indicating relatively limited explanatory power compared to more market-sensitive stocks.

The Adjusted R^2 of 0.2292, though lower, remains positive and confirms that the model retains explanatory relevance despite the small sample size of 19 observations. The slope coefficient of 2.67 is positive and statistically significant ($p \approx 0.02$), suggesting that a one-unit increase in Maruti's stock price is associated with an average increase of about 2.7 units in the Nifty 50. However, the magnitude of this effect is modest, highlighting Maruti's lower influence on daily index movements.

The F-statistic (6.35) with a significant p-value confirms that the regression model is statistically valid. Residuals fluctuate around zero but show relatively larger dispersion, indicating that a substantial portion of index movements is driven by other stocks and macroeconomic factors not captured by Maruti alone. Overall, the results suggest that Maruti Suzuki is less market-driven on a daily basis, reinforcing its relatively defensive and diversified behavior compared to more index-aligned stocks.

Mahindra & Mahindra

Regression Statistics					
Multiple R		0.583831312			
R Square		0.340859001			
Adjusted R Square		0.302086001			
Standard Error		457.8428301			
Observations		19			
ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	1842799.974	1842799.974	8.791143365	0.00867835
Residual	17	3563540.97	209620.0571		
Total	18	5406340.944			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	9382.945	4543.762	2.065	0.055	-203.555	18969.445	-203.555	18969.445
2585.1	4.956	1.672	2.965	0.009	1.429	8.483	1.429	8.483

RESIDUAL OUTPUT		
<i>Observation</i>	<i>Predicted 22124.7</i>	<i>Residuals</i>
1	22327.66864	-208.368645
2	22334.85493	-252.2049267
3	22896.12831	-558.8283069
4	22974.434	-429.733997
5	22902.32338	-349.8233773
6	22777.18295	-316.8829548
7	22494.68774	3.21225645
8	22532.84938	-62.34937735
9	22484.28003	-87.08002524
10	22789.8209	-281.0708984
11	23217.28076	-382.9807576
12	23205.88183	-298.281828
13	23399.16803	-208.5180251
14	23269.07155	81.32845371
15	23134.51462	523.8353833
16	22942.71524	725.9347635
17	22973.69059	513.1594114
18	22927.84707	664.1029326
19	22594.80008	924.5499184

Interpretation

The daily regression analysis between Mahindra & Mahindra and the Nifty 50 indicates a moderate and statistically significant linear relationship. The Multiple R value of 0.5838 suggests a moderate degree of co-movement between Mahindra & Mahindra's stock price and the index. The R^2 value of 0.3409 implies that approximately 34% of the variation in the Nifty 50 is explained by movements in Mahindra & Mahindra, indicating moderate explanatory power.

The Adjusted R^2 of 0.3021, which is slightly lower than R^2 , confirms that the model remains reliable despite the limited sample size of 19 observations. The slope coefficient of 4.96 is positive and statistically significant ($p \approx 0.009$), indicating that a one-unit increase in Mahindra & Mahindra's stock price leads to an average increase of about 5 units in the Nifty 50. This suggests that Mahindra & Mahindra has a meaningful but not dominant influence on daily index movements.

The F-statistic (8.79) with a significant p-value confirms the overall validity of the regression model. Residuals are distributed around zero, indicating no systematic prediction bias, though their dispersion highlights the influence of other market-wide and firm-specific factors beyond Mahindra & Mahindra.

Overall, the results suggest that Mahindra & Mahindra is moderately market-linked, offering some diversification benefits compared to more index-sensitive stocks.

Monthly price

TATA Motors

Regression Statistics	
Multiple R	0.809070734
R Square	0.654595452
Adjusted R Square	0.638147616
Standard Error	1412.447279
Observations	23

ANOVA					
	df	SS	MS	F	Significance F
Regression	1	79397845.68	79397845.68	39.79827296	2.95549E-06
Residual	21	41895153.62	1995007.315		
Total	22	121292999.3			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	14130.589	1283.171	11.012	0.000	11462.089	16799.090	11462.089	16799.090
424.25	9.877	1.566	6.309	0.000	6.621	13.133	6.621	13.133

RESIDUAL OUTPUT		
Observation	Predicted 17398.05	Residuals
1	18874.19033	-726.5403341
2	19416.95112	-929.2011168
3	19966.13218	-643.5821817
4	20488.15051	-754.6005141
5	20167.63027	-732.3302703
6	20258.99583	-730.2458252
7	20330.60667	-1341.456665
8	21098.57119	-830.6711942
9	21939.62817	-197.7281669
10	22807.84787	-1110.397872
11	23784.71851	-1445.968507
12	23931.39726	-1469.397263
13	24282.53731	-1634.337314
14	23518.52373	-254.6237285
15	24028.19536	113.7546357
16	25434.23717	-423.3371736
17	24923.08393	355.6160658
18	23664.21475	2132.685252
19	22461.6465	1842.703501
20	21934.19562	2341.85438
21	21531.19944	2211.700557
22	21109.43629	2372.713713
23	20265.90998	1853.390025

Interpretation

The monthly regression analysis of TATA Motors in relation to the Nifty 50 indicates a strong and statistically significant relationship at the monthly frequency. The Multiple R value of 0.809 reflects a high degree of linear association between TATA Motors' stock price and the index. The R^2 value of 0.655 shows that approximately 65.5% of the variation in TATA Motors' monthly prices is explained by movements in the Nifty 50, indicating substantial explanatory power. This is further supported by the Adjusted R^2 of 0.638, which remains close to R^2 and confirms the robustness of the model despite a relatively small sample size of 23 observations.

The slope coefficient ($\beta \approx 9.88$) is positive and highly statistically significant ($p < 0.001$), suggesting that a one-unit increase in the Nifty 50 leads to an average increase of about 9.9 units in TATA Motors' monthly stock price. This highlights TATA Motors' strong sensitivity to broader market movements at the monthly level. The intercept term is also statistically significant, indicating a stable baseline level of the stock price independent of index movements.

The F-statistic (39.80) with a very low significance value confirms that the overall regression model is statistically valid. Although residuals show noticeable dispersion, indicating the influence of firm-specific and sectoral factors, the model clearly demonstrates that market movements play a dominant role in explaining TATA Motors' monthly price behavior. Overall, the results suggest that TATA Motors is highly market-driven on a monthly basis, with limited diversification benefits when paired with the Nifty 50.

Maruti Suzuki

Regression Statistics					
Multiple R	0.85120678				
R Square	0.724552983				
Adjusted R Square	0.711436458				
Standard Error	1261.325736				
Observations	23				
ANOVA					
	df	SS	MS	F	Significance F
Regression	1	87883204.43	87883204.43	55.2397074	2.62482E-07
Residual	21	33409794.87	1590942.613		
Total	22	121292999.3			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	4954.419	2309.731	2.145	0.044	151.069	9757.768	151.069	9757.768
8503.05	1.516	0.204	7.432	0.000	1.092	1.941	1.092	1.941

RESIDUAL OUTPUT		
Observation	Predicted 17398.05	Residuals
1	18263.22502	-115.5750238
2	19099.4181	-611.6681002
3	19622.10511	-299.5551135
4	19688.44557	45.10442571
5	20620.24473	-1184.944732
6	20643.97566	-1115.225663
7	20463.30216	-1474.152156
8	21006.08085	-738.1808519
9	20547.53559	1194.364413
10	21085.31032	612.1396836
11	22575.8857	-237.1356981
12	24014.37434	-1552.374341
13	24371.85465	-1723.654652
14	23872.67112	-608.7711165
15	23315.41125	826.538754
16	25211.45952	-200.5595239
17	23798.74889	1479.951111
18	24918.7275	878.1724979
19	21801.10493	2503.245066
20	21997.16943	2278.880573
21	21950.16247	1792.737528
22	24540.39733	-1058.247331
23	22810.38975	-691.0897497

Interpretation

The monthly regression analysis of Maruti Suzuki in relation to the Nifty 50 indicates a strong and statistically significant linear relationship at the monthly frequency. The Multiple R value of 0.851 reflects a high degree of linear association between Maruti Suzuki's stock price and the index. The R^2 value of 0.725 shows that approximately 72.5% of the variation in Maruti Suzuki's monthly prices is explained by movements in the Nifty 50, indicating strong explanatory power. This is further supported by the Adjusted R^2 of 0.711, which remains close to R^2 and confirms the robustness of the model across 23 observations.

The slope coefficient ($\beta \approx 1.52$) is positive and highly statistically significant ($p < 0.001$), indicating that a one-unit increase in the Nifty 50 leads to an average increase of about 1.5 units in Maruti Suzuki's monthly stock price. This suggests that Maruti Suzuki is highly responsive to broader market movements at the monthly level. The intercept term is also statistically significant, indicating the presence of a stable price component independent of index movements.

The F-statistic value of 55.24, along with a very low significance level, confirms that the regression model is statistically valid. Although the residuals display some dispersion—highlighting the influence

of firm-specific and operational factors—the results clearly demonstrate that market movements are the dominant driver of Maruti Suzuki’s monthly price behavior. Overall, the findings suggest that Maruti Suzuki is strongly market-aligned on a monthly basis, offering limited diversification benefits when combined with Nifty 50 exposure.

Mahindra & Mahindra

Regression Statistics					
Multiple R		0.925284189			
R Square		0.85615083			
Adjusted R Square		0.849300869			
Standard Error		911.5109723			
Observations		23			
ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	103845102	103845102	124.9862435	2.65639E-10
Residual	21	17447897.31	830852.2527		
Total	22	121292999.3			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	15056.829	650.293	23.154	0.000	13704.470	16409.188	13704.470	16409.188
1169.9	3.196	0.286	11.180	0.000	2.601	3.790	2.601	3.790

RESIDUAL OUTPUT		
Observation	Predicted 17398.05	Residuals
1	19010.75066	-863.1006559
2	19271.51199	-783.7619944
3	19747.81687	-425.2668658
4	19828.98522	-95.43522358
5	20144.07184	-708.7718409
6	19969.75161	-441.0016079
7	19705.15554	-716.0055438
8	20251.28438	16.61562343
9	20499.90242	1241.997583
10	20333.25164	1364.198364
11	21361.59721	977.1527917
12	21179.44774	1282.552256
13	22039.70451	608.4954874
14	23474.05165	-210.1516546
15	24246.90864	-104.9586365
16	24095.2772	915.6228036
17	23931.02312	1347.676882
18	25172.51552	624.38448
19	24060.92445	243.4255535
20	24696.05087	-420.0008684

21	24905.68253	-1162.782533
22	24888.90561	-1406.755608
23	23403.42879	-1284.128792

Interpretation

The monthly regression analysis of Mahindra & Mahindra in relation to the Nifty 50 indicates a very strong and statistically significant linear relationship at the monthly frequency. The Multiple R value of 0.925 reflects a very high degree of linear association between Mahindra & Mahindra's stock price and the index. The R^2 value of 0.856 shows that approximately 85.6% of the variation in Mahindra & Mahindra's monthly prices is explained by movements in the Nifty 50, indicating excellent explanatory power. This is further supported by the Adjusted R^2 of 0.849, which remains close to R^2 and confirms the robustness of the model across 23 observations.

The slope coefficient ($\beta \approx 3.20$) is positive and highly statistically significant ($p < 0.001$), indicating that a one-unit increase in the Nifty 50 results in an average increase of about 3.2 units in Mahindra & Mahindra's monthly stock price. This highlights Mahindra & Mahindra's strong sensitivity to broader market movements at the monthly level. The intercept term is also statistically significant, suggesting a stable base level of the stock price independent of index fluctuations.

The F-statistic value of 124.99, along with an extremely low significance level, confirms that the regression model is statistically valid and highly reliable. Although the residuals show some dispersion—indicating the influence of firm-specific and sectoral factors—the results clearly demonstrate that market movements are the dominant driver of Mahindra & Mahindra's monthly price behavior. Overall, the findings suggest that Mahindra & Mahindra is highly market-driven on a monthly basis, offering minimal diversification benefits when combined with Nifty 50 exposure.

Yearly price

TATA Motors

Regression Statistics	
Multiple R	0.798621112
R Square	0.637795681
Adjusted R Square	0.586052206
Standard Error	3582.189473
Observations	9

ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	158169628.6	158169628.6	12.32610856	0.009848034
Residual	7	89824569.92	12832081.42		
Total	8	247994198.6			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	7890.686	2349.050	3.359	0.012	2336.064	13445.307	2336.064	13445.307
379.65	16.381	4.666	3.511	0.010	5.348	27.414	5.348	27.414

RESIDUAL OUTPUT		
Observation	Predicted 7713.05	Residuals
1	15596.26936	-6358.419356
2	13446.27391	-3234.473913
3	10959.65061	709.499387
4	9003.769045	-749.9690448
5	12931.9131	1935.436901
6	15117.12751	2553.322486
7	14840.29001	2557.759994
8	24144.65124	-1682.651235
9	18896.20522	4269.494782

Interpretation

The yearly regression analysis of TATA Motors in relation to the Nifty 50 indicates a strong and statistically significant linear relationship over the long-term horizon. The Multiple R value of 0.799 suggests a high degree of linear association between TATA Motors' yearly stock price movements and the Nifty 50. The R^2 value of 0.638 indicates that approximately 63.8% of the variation in TATA Motors' yearly prices is explained by movements in the Nifty 50, demonstrating substantial explanatory power even at a low-frequency (annual) level.

The Adjusted R^2 of 0.586, though slightly lower, remains reasonably close to R^2 , confirming the stability of the model despite the small sample size of 9 observations. The slope coefficient ($\beta \approx 16.38$) is positive and statistically significant ($p \approx 0.01$), indicating that a one-unit increase in the Nifty 50 leads to an average increase of about 16.4 units in TATA Motors' yearly stock price. This highlights

TATA Motors' strong long-term sensitivity to overall market movements.

The F-statistic (12.33) with a significant p-value confirms that the regression model is statistically valid. The residuals show noticeable dispersion, reflecting the influence of firm-specific, sectoral, and macroeconomic factors not fully captured by the index. Overall, the results suggest that TATA Motors is significantly market-driven even on a yearly basis, with Nifty 50 movements acting as a key determinant of its long-term price behavior, though some diversification potential remains due to company-specific influences.

Maruti Suzuki

Regression Statistics	
Multiple R	0.798621112
R Square	0.637795681
Adjusted R Square	0.586052206
Standard Error	3582.189473
Observations	9

ANOVA	df	SS	MS	F	Significance F
Regression	1	158169628.6	158169628.6	12.32610856	0.009848034
Residual	7	89824569.92	12832081.42		
Total	8	247994198.6			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	7890.686	2349.050	3.359	0.012	2336.064	13445.307	2336.064	13445.307
379.65	16.381	4.666	3.511	0.010	5.348	27.414	5.348	27.414

RESIDUAL OUTPUT		
Observation	Predicted 7713.05	Residuals
1	15596.26936	-6358.419356
2	13446.27391	-3234.473913
3	10959.65061	709.499387
4	9003.769045	-749.9690448
5	12931.9131	1935.436901
6	15117.12751	2553.322486
7	14840.29001	2557.759994
8	24144.65124	-1682.651235
9	18896.20522	4269.494782

Interpretation

The yearly regression analysis of Maruti Suzuki in relation to the Nifty 50 indicates a strong and statistically significant linear relationship over the long-term horizon. The Multiple R value of 0.799

reflects a high degree of linear association between Maruti Suzuki's yearly stock price movements and the Nifty 50. The R^2 value of 0.638 suggests that approximately 63.8% of the variation in Maruti Suzuki's yearly prices is explained by movements in the Nifty 50, indicating substantial explanatory power at the annual level.

The Adjusted R^2 of 0.586, although lower than R^2 , remains reasonably strong and confirms the reliability of the regression model despite the small sample size of 9 observations. The slope coefficient ($\beta \approx 16.38$) is positive and statistically significant ($p \approx 0.01$), indicating that a one-unit increase in the Nifty 50 leads to an average increase of about 16.4 units in Maruti Suzuki's yearly stock price. This highlights Maruti Suzuki's strong sensitivity to long-term market movements.

The F-statistic (12.33) with a significant p-value confirms the overall validity of the regression model. The residuals exhibit noticeable dispersion, suggesting the presence of firm-specific, competitive, and macroeconomic factors influencing Maruti Suzuki's performance beyond general market trends. Overall, the results indicate that Maruti Suzuki is significantly market-driven on a yearly basis, though company-specific fundamentals continue to play a meaningful role in shaping long-term returns.

Mahindra & Mahindra

Regression Statistics	
Multiple R	0.797677874
R Square	0.636289991
Adjusted R Square	0.584331419
Standard Error	3589.627361
Observations	9

ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	157796226.4	157796226.4	12.2461022	0.010000541
Residual	7	90197972.14	12885424.59		
Total	8	247994198.6			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	7969.840	2336.536	3.411	0.011	2444.811	13494.869	2444.811	13494.869
1195.2	6.121	1.749	3.499	0.010	1.985	10.257	1.985	10.257

RESIDUAL OUTPUT		
<i>Observation</i>	<i>Predicted 7713.05</i>	<i>Residuals</i>
1	15844.62108	-6606.771077
2	12548.72073	-2336.920731
3	12015.26783	-346.1178287
4	9639.979261	-1386.179261
5	12913.23158	1954.118422
6	13029.22621	4641.223786
7	15130.90211	2267.147895
8	19697.54052	2764.459482
9	24116.66069	-950.9606871

Interpretation

The yearly regression analysis of Mahindra & Mahindra in relation to the Nifty 50 shows a strong and statistically significant linear relationship over the long-term horizon. The Multiple R value of 0.798 indicates a high degree of linear association between Mahindra & Mahindra's yearly stock price movements and the Nifty 50. The R^2 value of 0.636 suggests that approximately 63.6% of the variation in Mahindra & Mahindra's yearly prices is explained by movements in the Nifty 50, reflecting substantial explanatory power at the annual level.

The Adjusted R^2 of 0.584, although slightly lower than R^2 , remains reasonably strong and confirms the reliability of the regression model despite the limited sample size of 9 observations. The slope coefficient ($\beta \approx 6.12$) is positive and statistically significant ($p \approx 0.01$), indicating that a one-unit increase in the Nifty 50 leads to an average increase of about 6.1 units in Mahindra & Mahindra's yearly stock price. This highlights Mahindra & Mahindra's meaningful sensitivity to long-term market movements, though at a lower magnitude compared to some peers.

The F-statistic (12.25) with a significant p-value confirms the overall validity of the regression model. The residuals show noticeable dispersion, indicating the influence of firm-specific, sectoral, and macroeconomic factors beyond general market trends. Overall, the findings suggest that Mahindra & Mahindra is significantly market-driven on a yearly basis, while still retaining some scope for diversification due to company-specific performance drivers.

Overall Interpretation

1. Performance in Terms of Returns and Stability

The return analysis across daily, monthly, and yearly periods shows clear differences among the selected automobile companies. TATA Motors delivered the highest return potential, especially over monthly and yearly horizons, but these higher returns were accompanied by significant volatility. This indicates strong growth potential, but also higher risk and price instability.

In contrast, Maruti Suzuki emerged as the most stable performer. Although its average returns were lower, the stock showed consistent performance with comparatively low volatility, particularly in the long term. This stability reflects Maruti's strong market position and defensive business model.

Mahindra & Mahindra displayed a balanced risk–return profile, offering moderate-to-high returns with controlled volatility. Its diversified business across automobiles and rural segments helped reduce extreme fluctuations, placing it between TATA Motors and Maruti Suzuki in terms of risk and return.

2. Comparison with the NSE Index in Terms of Risk and Return

Compared to the Nifty 50 Index, all three automobile stocks offered higher return potential but carried greater risk. The Nifty 50 delivered smoother and more stable returns, reinforcing its role as a diversified benchmark.

Among the stocks, TATA Motors showed the strongest linkage with the index, indicating high exposure to market-wide movements. Maruti Suzuki exhibited lower short-term dependence but strong long-term alignment with the index, while Mahindra & Mahindra showed moderate market sensitivity, benefiting from business diversification. Overall, automobile stocks tend to outperform the index in bullish markets but underperform more sharply during downturns.

3. Practical Implications for Investors

From an investment perspective, Maruti Suzuki and Mahindra & Mahindra offer better diversification benefits, particularly in short-term portfolios, while TATA Motors suits investors with higher risk appetite seeking higher returns. The findings also highlight that diversification benefits are stronger in the short term and reduce over longer horizons, making investment horizon an important factor.

A balanced portfolio combining the Nifty 50 with selective automobile stocks, such as Maruti for stability and M&M for balance, can help investors achieve an optimal mix of growth and risk control.

Conclusion

This study provides a clear understanding of how selected automobile stocks - TATA Motors, Maruti Suzuki, and Mahindra & Mahindra behave in relation to the Nifty 50 Index across different time horizons. The analysis highlights that while automobile stocks can deliver higher returns than the broader market, they also involve greater levels of risk and price volatility. The relationship between stock performance and the market is not uniform and varies significantly depending on the company's business model and the investment horizon considered.

Among the three companies, TATA Motors stands out as a high-growth stock, offering strong return potential but accompanied by considerable volatility. Its performance is closely linked to market movements, making it more suitable for investors with a higher risk appetite who are comfortable with short-term fluctuations. Maruti Suzuki, in contrast, demonstrates stability and consistency, reflecting its strong market leadership and predictable operational performance. This makes Maruti Suzuki a preferable choice for conservative investors seeking steady long-term returns with lower risk. Mahindra & Mahindra occupies a middle ground, offering a balanced combination of growth and stability due to its diversified presence across multiple segments.

The comparison with the Nifty 50 Index further reinforces the importance of diversification. While individual stocks may outperform the index during favourable market conditions, the index provides smoother and more stable returns due to its diversified composition. This highlights that relying solely on individual stocks can increase portfolio risk, whereas combining them with index exposure can improve overall stability.

From an investor's perspective, the findings emphasise that there is no single "best" stock, but rather the right choice depends on risk tolerance, return expectations, and investment horizon. Short-term investors benefit more from diversification, while long-term investors should focus on fundamentals and market trends. Overall, the study underlines the importance of thoughtful stock selection, diversification, and horizon-based investment planning in achieving sustainable investment outcomes.

References

1. **National Stock Exchange of India.** (2024). *Historical equity and index price data.*
2. **CFA Institute.** (2023). *Correlation analysis.*
<https://www.cfainstitute.org/en/membership/professional-development/refresher-readings/correlation-analysis>
3. **CFA Institute.** (2023). *Linear regression.*
<https://www.cfainstitute.org/en/membership/professional-development/refresher-readings/linear-regression>
4. **Damodaran, A.** (2023). *Risk and return models.* NYU Stern School of Business.
https://pages.stern.nyu.edu/~adamodar/New_Home_Page/finrisk.htm
5. **Gujarati, D. N., & Porter, D. C.** (2009). *Basic econometrics* (5th ed.). McGraw-Hill Education.
<https://www.mheducation.com/highered/product/basic-econometrics-gujarati/M9780073375779.html>
6. **Markowitz, H.** (1952). Portfolio selection. *The Journal of Finance*, 7(1), 77–91.
7. **Microsoft Corporation.** (2024). *CORREL function in Excel.*
8. **Microsoft Corporation.** (2024). *Use the Analysis ToolPak for regression analysis.*
<https://support.microsoft.com/en-us/office/use-the-analysis-toolpak>

Excel Link-

[FM CEC1 CompanyAnalysis AutomobileSector.xlsx](#)